VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI – 590018

A project Phase I Report on

LiFi Based Automatic Traffic Signal Control For Ambulance

Submitted in partial fulfillment of the requirements for the VII Semester of degree of **Bachelor of Engineering in Information Science and Engineering** of Visvesvaraya Technological University, Belagavi

Submitted By

Amardeep Kumar	**1RN19IS020**
Amartya	**1RN19IS021**
Anushka Jaiswal	**1RN19IS035**
Devendra Upadhyay	**1RN19IS056**

Under the Guidance of

Ms. Vanishri S

Assistant Professor

Department of ISE

ESTD:2001

An Institute with a Difference

Department of Information Science and Engineering

RNS Institute of Technology

Dr. Vishnuvaradhan Road, Rajarajeshwari Nagar post, Channasandra, Bengaluru-560098

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RNS INSTITUTE OF TECHNOLOGY

Dr. Vishnuvaradhan Road, Rajarajeshwari Nagar post,

Channasandra, Bengaluru - 560098

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

CERTIFICATE

Certified that the project work phase 1 entitled **LiFi based automatic traffic signal control for ambulance** has been successfully completed by **Amardeep Kumar (1RN19IS020), Amartya (1RN19IS021), Anushka Jaiswal (1RN19IS035) and Devendra Upadhyay (1RN19IS056)** bona fide students of **RNS Institute of Technology, Bengaluru** in partial fulfillment of the requirements for the award of degree in **Bachelor of Engineering in Information Science and Engineering** of **Visvesvaraya Technological University, Belagavi** during academic year **2022-2023**. The project phase 1 report has been approved as it satisfies the academic requirements in respect of project phase 1 work for the said degree.

______________________	______________________	______________________
Ms. Vanishri S	**Dr. Suresh L**	**Dr. M K Venkatesha**
Project Guide	Professor and HoD	Principal
Assistant Professor	Department of ISE	RNSIT
Department of ISE	RNSIT	

External Viva

Name of the Examiners	**Signature with Date**
1. ______________________	**1.** ______________________
2. ______________________	**2.** ______________________

DECLARATION

We, **AMARDEEP KUMAR [1RN19IS020], AMARTYA [1RN19IS021], ANUSHKA JAISWAL [1RN19IS035] and DEVENDRA UPADHYAY [1RN19IS056]** students of VII Semester BE, in **Information Science and Engineering**, **RNS Institute of Technology** hereby declare that the Project work phase 1 entitled ***LiFi based automatic traffic signal control for ambulance*** has been carried out by us and submitted in partial fulfillment of the requirements for the *VII Semester degree of* ***Bachelor of Engineering in Information Science and Engineering*** *of* ***Visvesvaraya Technological University, Belgaum*** during academic year **2022-2023**.

Place: Bengaluru

Date:

AMARDEEP KUMAR [1RN19IS020]
AMARTYA [1RN19IS021]
ANUSHKA JAISWAL [1RN19IS035]
DEVENDRA UPADHYAY [1RN19IS056]

ABSTRACT

India is one of the fastest developing country in the world. One of the disadvantages of this unprecedented growth is the ineffective traffic management. Road traffic congestion becomes a major issues for highly crowded metropolitan cities like, Bengaluru. Ambulance service is one of the major services which gets affected by traffic jams. Ambulance needs to be on time as it is life deciding factors in emergency cases. In India, many lives have been lost as emergency vehicles like ambulance or firefighting vehicles were not able to reach their destination on time due to heavy traffic. An efficient and feasible traffic management system can be implemented using Li-Fi.

Light Fidelity (Li-Fi) is a form of bi-directional Visible Light Communication (VLC) in which light is modulated at speeds greater than a human eye can follow. Li-Fi can be used to implement Vehicle to Vehicle (V2V) communication as it has many advantages over other communication protocols. One of the main advantages of Li-Fi is that it provides connectivity within a very large area with more security and higher data rates. Emergency vehicles such as Ambulances, Firefighting trucks, Police vehicles etc. can propagate faster through traffic-dense roads using Li-Fi based V2V communication system.

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Place: Bengaluru
Date:

AMARDEEP KUMAR [1RN19IS020]
AMARTYA [1RN19IS021]
ANUSHKA JAISWAL [1RN19IS035]
DEVENDRA UPADHYAY [1RN19IS056]

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Chapter 1

INTRODUCTION

1.1 Domain background

The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from ordinary household objects to sophisticated industrial tools. With more than 7 billion connected IoT devices today, experts are expecting this number to grow to 10 billion by 2020 and 22 billion by 2025. Oracle has a network of device partners. Over the past few years, IoT has become one of the most important technologies of the 21st century. Now that we can connect everyday objects—kitchen appliances, cars, thermostats, baby monitors—to the internet via embedded devices, seamless communication is possible between people, processes, and things.

1.2 Project Overview

Wireless communication has been solely based on radio waves. Due to technological developments, this spectrum has become congested. Similarly, the traffic density has also increased many folds. To have an efficient traffic management, we need reliable wireless V2V communication system. The choice of radio waves for this application involves a huge bandwidth to be allocated to implement V2V communication system, which is very difficult as this spectrum is already congested due to human cellular communication systems. Also, if we consider the circuitry associated with this type of communication, it is expensive. Therefore, the need for an alternative spectrum is increasing day by day. Gamma rays, X-rays, Infra-red and Ultraviolet rays cannot be used to implement V2V communication system due to their harmful effects on human body. VLC is an emerging communication platform which uses visible light in the frequency range of 400 to 800 THz. Li-Fi is a VLC protocol, which is like the Wi-Fi protocol in every layer except the physical layer. In India, many lives have been lost as emergency vehicles like ambulance or firefighting vehicles were not able to reach their destination on time due to heavy traffic. An efficient and feasible traffic management system can be implemented using Li-Fi.

Chapter 2

LITERATURE SUVERY

A literature survey or a literature review in a project report shows the various analyses and research made in the field of interest and the results already published, taking into account the various parameters of the project and the extent of the project. Literature survey is mainly carried out in order to analyze the background of the current project which helps to find out flaws in the existing system & guides on which unsolved problems we can work out. So, the following topics not only illustrate the background of the project but also uncover the problems and flaws which motivated to propose solutions and work on this project.

A literature survey is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews use secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such as a thesis, dissertation or a peer-reviewed journal article, a literature review usually precedes the methodology and results sectional though this is not always the case. Literature reviews are also common in are search proposal or prospectus (the document that is approved before a student formally begins a dissertation or thesis). Its main goals are to situate the current study within the body of literature and to provide context for the particular reader. Literature reviews are a basis for researching nearly every academic field. demic field. A literature survey includes the following:

- Existing theories about the topic which are accepted universally.
- Books written on the topic, both generic and specific.
- Research done in the field usually in the order of oldest to latest.
- Challenges being faced and on-going work, if available.

Literature survey describes about the existing work on the given project. It deals with the problem associated with the existing system and also gives user a clear knowledge on how to deal with the existing problems and how to provide solution to the existing problems.

Objectives of Literature Survey

- Learning the definitions of the concepts.
- Access to latest approaches, methods and theories.
- Discovering research topics based on the existing research
- Concentrate on your own field of expertise– Even if another field uses the same words, they usually mean completely.
- It improves the quality of the literature survey to exclude sidetracks– Remember to explicate what is excluded.

Before building our application, the following system is taken into consideration:

[1] Modeling the Impact of VANET-Enabled Traffic Lights Control on the Response Time of Emergency Vehicles in Realistic Large-Scale Urban Area, 2013 (Hamed Noori)

The contribution of this paper is twofold: First, the effect of the changing traffic lights status to green for emergency cars is investigated by using traffic simulator (SUMO). Second, this paper uses OMNET++ (Network Simulator) in order to simulate the mentioned scenario as a VANET (with 802.11p standard) by using Veins framework to run SUMO and OMNET++ in parallel. This study has developed the Veins framework by adding a new module to OMNET++ to consider the traffic lights which are simulated in SUMO. Moreover this study has developed a new program written in Python which is connected to SUMO and controls the traffic simulation. This program uses SUMO to simulate a microscopic traffic (by considering every single vehicle movements) and also a city with intelligent traffic lights. Additionally, several statistics about traffic simulation is created for each car such as traveling time, waiting time, emissions, fuel consumption; or complete amount of car emissions in the street during the simulation, fuel consumption, number of vehicles and so on, for each street. Additionally, several statistics about traffic simulation is created for each car such as traveling time, waiting time, emissions, fuel consumption; or complete amount of car emissions in the street during the simulation, fuel consumption, number of vehicles and so on, for each street. This paper uses Manhattan realistic map to describe the mentioned program, then uses realistic map and realistic traffic demand of Cologne, Germany, to obtain a realistic and reliable result.

[2] A GPS Based Traffic Light Pre-emption Control System for Emergency Vehicles, 2013 (Abubakr S. Eltayeb; Halla O. Almubarak; Tahani Abdalla Attia)

In this paper, the design and implementation of an automatic preemption traffic control system, based on the global positioning system satellites. It ensures the arrival of emergency vehicles to their destinations on the minimum time possible. The results indicate that the proposed system has an optimum solution for the delay time experienced on the emergency vehicles along their path. In the simulation program ISIS professional and an Atmega16 microcontroller was programmed to act and response as GSM modem proposed system is giving a unique and optimum solution to the main problem and other problems attached to the different pre-emption systems by suggesting the shortest path with the least obstacles taken from an updated electronic map to the driver of the emergency vehicle, which will result in reducing the emergency vehicles' travelling time to the desired destination and clearing the path in advance from other vehicles and pedestrians. Field experiment tests on the hardware model on Othman Digna Street across Al-Jami'a Street at the city of Khartoum were successfully accomplished with fair results proved to be compatible with theoretic results to a great extend.

[3] Implementing Intelligent Traffic Control System for Congestion Control, Ambulance Clearance and Stolen Vehicle Detection, 2015 (Rajeshwari S., Santhoshs Hebbar, Varaprasad Golla)

This paper presents an intelligent traffic control system to pass emergency vehicles smoothly. Each individual vehicle is equipped with special RFID tag(placed at a strategic location), which makes it impossible to remove or destroy. We use RFID reader, NSK EDK-125–TTL and PIC16F877A system-onchip to read the RFID tags attached to the vehicle. It counts number of vehicles that passes on a particular path during a specified duration. It also determines the network congestion, and hence the green light duration for that path. If the RFID-tag-read belongs to the stolen vehicle, then a message is sent using GSM SIM300 to the police control room. Also, when an ambulance is approaching the junction, it will communicate to the traffic controller in the junction to turn on the green light. This module uses ZigBee modules on CC2500 and PIC16F877A system-onchip for wireless communications between the ambulance and traffic controller. The prototype was tested under different combinations of inputs in our wireless communication laboratory and experimental results were found as expected.

[4] GPS and ZigBee Based Traffic Signal Preemption , 2016 (Vineeth Kodire, Sreebha Bhaskaran, Vishwas H.N.)

In this paper, whenever the emergency vehicle is approaching preemption messages are provided to the traffic signal unit. So these signals interrupt the normal operation of the traffic light and make it green for the emergency vehicle. It is a type of Vehicle-to-Infrastructure (V2I) communication. The proposed system uses GPS method to get the location data and employs ZigBee for communicating efficiently with the intersection module. There is no requirement of Line-ofSight employed in infrared based preemption methods. Also eliminate the false trigger occurring in sound (siren) based method due to noise and fading. Using this approach, it is possible to serve multiple emergency vehicles reaching the traffic intersection according to their distance from the intersection and accurately taking the decision to turn the signal green for the particular vehicle. There is no involvement of driver of the EV in anyway, it is fully autonomous and it can be used as a part of ITS.

[5] Advanced Automation Control in an Ambulance under Emergency Condition , 2017 (Leila Sai Krishna; Samineni Vijay Chowdary; M. Pushpavalli; P. Sivagami)

In this paper, they have implemented a scheme which can control the traffic signals automatically in its path way and reduce the amount of traffic congestion at the signal. The ambulance is implemented with embedded system units, which finds the accident spot and delivers the spot to the close by ambulance through GPS .The ambulance guides the traffic lights in the roadway to the hospital and it also checks the next consecutive signal to shorten the time loss. The vehicle unit is equipped with vibration sensor to determine the vibration if it exceeds the level .The embedded unit sense the accident happens and its sends the neighborhood ambulance unit through wireless transmission. RFID reader is needed to extricate the ambulance from other vehicles.

[6] Emergent vehicle tracking system using IR sensor, 2017 (Manjunath Managuli; Abhaya Deshpande; Sudha H Ayatti)

In this paper, they proposed a signal track division that use universal position method to find the exact position of an emergency van (ex. Ambulance, Fire exchange) and remote users get the information transmitted using global position system which also provides traffic signal track monitoring system i.e., during object motion IR sensors detect the ambulance, pass the message to main controller which reflects the LED bulb to glow leading to closing of all the signals track, and immediately the emergent vehicle pass the signal track. The main principle of this scheme is to plan an emergent tracking method. The proposed system is implemented based on real time embedded system for this implementation they used ATMEG328P controller which is the heart of the system. The IR sensor is used to sense the data which is processed by the controller and sent the data to emergent tracking system through GPS.

[7] Intelligent Ambulance with Automatic Traffic Control, 2020 (P. Devi; S. Anila)

This paper have come up with the solution of “Intelligent automatic traffic control for ambulance ”. The proposed system creates a android app that connects both the ambulance and the traffic signal station using cloud network. This system makes uses RFID (radio frequency identification) technology to implement the Intelligent traffic signal control. The basic idea behind the proposed system is, if the Ambulance halts on the way due to a traffic signal, RFID installed at the traffic signal tracks the RFID tagged ambulance and sends the data to the cloud. After the acknowledgment for the user through the mobile app, the particular signal is made Green for some time and after the ambulance passes by, it regains its original flow of sequence of signaling If, this scheme is fully automated, it finds the ambulance spot, controls the traffic lights. This system control the traffic lights and save the time in emergency periods.

[8] Leveraging Emergency Response System Using the Internet of Things, 2020 (Luz A. Magre Colorado; Jorge Franco Ibañez; Juan Carlos Martinez-Santos)

This paper propose a hardware/software system that allows the emergency vehicles to “talk” to the traffic-lights nearby to prioritize them over the other cars while maintaining the roads safe. This system should take the current infrastructure (surveillance camera network and the traffic-light system) to coordinate the best route. This system should connect the surveillance camera network and the traffic-light system to coordinate the best route. Our main objective is to develop a hardware/software system that allows the emergency vehicles to “talk” with the traffic-lights around to prioritize them over the other cars while keeping the safety on the roads. This system should connect both the surveillance camera network and the trafficlight system to coordinate the best route.

The proposal is an Emergency Vehicle Signal Preemption System in cooperatives Vehicle Infrastructure System. However, the approach will complement the systems of intelligent traffic lights based on readings of transit through the checkpoints (cameras infrastructure system). Using the technology of the QR code, mobile networks, and GPS for the control and automation of the traffic lights in favor of the emergency vehicles that need to transit through high congestion road zones using the infrastructure explained. For most applications focusing on emergency management, RFID technology is the most used. They discarded its use in our solution because they are not communicating vehicles with each other in a network of cars. They prefer an IoT approach instead of using a wireless sensor network (WSN). In this way, they can remove the errors that the system may have because the tag could not detect a vehicle. After all, it was going too fast for the system to catch it. We use the GSM network to communicate any updates, GPRS to monitor the location of the vehicle, and checkpoints to confirm how the execution of the operation is going.

[9] Li-Fi technology in traffic light, 2017 (V.K.G. Kalaiselvi; A. Sangavi; Dhivya)

The proposed paper aims in using the Wi-Fi technology and enabling communication of vehicles with the traffic light system in order to prioritize the vehicles and change the signals accordingly rather than by a process of pre-defined order or by manual order. Traffic lights already use LED lighting, so that this proposed system may seem easy to implement. Sending data through siren lights in an ambulance and fire extinguishers to a traffic light control system and switching the signal in order to allow faster and non-interrupted transport. Here is already a kind of system in play for GPS navigational systems, but the traffic lights can be used in updating drivers using basic information or streaming video directly from news broadcasts i.e, about the accidents or delays up ahead.

[10] Efficiency of opportunistic cellular/LiFi traffic offloading, 2017 (Vishnu Raj RS,Athi Narayanan S,Kamal Bijlani)

In this paper, our main aim is to discuss the traffic offloading efficiency of opportunistic cellular/LiFi HetNet. In addition, the effect of LiFi-signal blocking is taken into consideration. Furthermore, the effects of user existence in LiFi cell and dynamic variation of user densities on LiFi offloading efficiency are considered. The rest of this paper is organized as follows. In Section 2, state- and timing-diagram for different offloading cases are explained and mathematical expressions for the distributions of different timing parameters are given. In

Section 3, the LiFi offloading efficiency is investigated numerically. We have investigated traffic offloading efficiency of cellular/LiFi HetNet. A framework of opportunistic cellular/LiFi offloading and timing diagrams for different disjoint cases have been illustrated. Statistical distributions for different times have been explored. Our results reveal that LiFi blocking reduces the traffic offloading efficiency. In addition, the users' density has a positive effect on LiFi offloading efficiency in opportunistic cellular/LiFi HetNet.

[11] Li-Fi Based Smart Traffic Network, 2018 (A. Correa; A. Hamid; E. Sparks)

In this paper, The Smart Traffic System will utilize Li-Fi to avoid using the already crowded radio air waves. Our system will also eliminate a problem faced by the Tesla Autopilot System, a currently popular autonomous car technology. This system is currently designed to be used on limited-access highways, and not in urban streets as it is unable to improvise in construction areas and locations with no lane markings. Our system would require a light transmitter be placed on each traffic sign to send a signal to cars that alert the system of the position of each sign so the car can determine the action required by each sign. As traffic signals already emit light, they would not require an additional transmitter. They would only need to be reprogrammed in order to change the flash rate of the existing lights. This project was simulated using Intel's Altera Cyclone V FPGA.

[12] Li-Fi Based Automatic Traffic Signal Control for Emergency Vehicles, 2018 (R Shanmughasundaram; S Prasanna Vadanan; Vivek Dharmarajan)

In this paper, In this paper, a V2V communication system is proposed in which the headlight and tail light of a vehicle are made as Li-Fi transmitter and Li-Fi receiver respectively. Using the proposed system, when a traffic signal receives an alert regarding an emergency vehicle, the traffic signal would immediately turn green. Thus, decreasing the waiting time of emergency vehicles in traffic dense lanes. In this paper, a single hop scenario was implemented. The time taken for the alert message to propagate from an emergency vehicle to a non-emergency vehicle and then to the traffic signal control is approximately one second. This time can be further reduced by implementing the same functionality using optimized code. The time can also be reduced by designing better modulation and demodulation circuits which supports higher data rates. The alert message can also be encrypted to make it secure.

[13] Highway Navigation Using Light Fidelity Technology, 2015 (Abhishek Patni, Bhavini Mishra, Harsh Aditya, Yogesh Kumar and Rohit Sharma)

In this paper, they presented light is everywhere street lamp, traffic light and meeting hall. Imagine that all these light bulbs are high speed wireless transmitter that connects either human to human or systems with systems. Light fidelity or we can say LIFI, a new WIFI is based on the very recent intelligence of light emitting diode (LED) technology as the source of lightning. LIFI is a fully networked, bidirectional high speed wireless connection system. Unlike wireless networks.

LIFI networks do not rely on any fixed frequency spectrum which is very limited. Instead, LIFI is based on visible light spectrum which provides the spectrum which is more than 10,000 times to radio wave to deliver desired data. These unique characteristics of LIFI pose a number of challenges for the implementation of capacity, efficiency, availability and security in the wireless network system. In this Topic, we study the LIFI use in highway navigation system thus understanding the vulnerabilities and how LIFI will use full in future. Further, we discuss the various issues in WIFI and explore approaches to solve the problems in communication.

[14] Intelligent traffic signal control system for ambulance using RFID and cloud, 2017 (J. Saradha, G. Vijayshri and T. Subha)

The main concept behind the paper is to provide a smooth flow for the ambulance to reach the hospitals in time and thereby minimizing the delay caused by traffic congestion. The Microcontroller based RFID system is used to alter the traffic lights upon its arrival at traffic light junction which would save a lives at critical time. Radio Frequency Identification is a tiny electronic gadgets that comprise of a small chip and an antenna. The small chip is

embedded with information's about patience's status and the ambulance current lane .The RFID reader located at the traffic signal reads these information from the RFID locator installed at the ambulance . To avoid unnecessary traffic

signal changes, we cross refer the ambulance current location and patience's condition using mobile app registered by the ambulance driver.. In case of network failure, the RFID takes the whole control.

[15] Li-Fi Communications in Smart Cities for Truly Connected Vehicles, 2021 (Jamshaid Iqbal Janjua; Tahir Abbas Khan; Muhammad Saeed Khan; Mehwish Nadeem)

Our paper speaks about the future communication of Li-Fi and its implementation in connected vehicles in smart cities. The term Li-Fi is used for high-speed VLC which offers a higher bandwidth, connectivity and secure communication between vehicle-to-vehicle communication. Besides the strengths of Li-Fi technology, the aspect of weaknesses of Li-Fi technology are also discussed and summarized. Vehicles and traffic infrastructure communications can gain benefits that are wide-ranging, by implementing Li-Fi technology as a communication medium. We have proposed a Li-Fi "Transmitter" and "Receiver" model that enable safe inter-operative network wireless communications between automotive and implementation architecture is presented for Li-Fi communication between automotive and traffic infrastructure for vehicle integration. This paper also demonstrates a VLC based communication system in which data is sent from Li-Fi transmitter to a Li-Fi receiver using visible light communication (VLC) as a medium.

Summary

Research on traffic management, V2V communication, Ambulance based traffic control has been carried out since 1990s. Various communication protocols like GSM, RFID, IR, ZigBee etc. have been considered to implement emergency vehicle based traffic control. However, the above methods cannot be used in today's world for various reasons.

GSM based communication a system has been extensively used for cellular communication. The infrastructure needed for GSM communication is high compared to other communication protocols.

Even though basic infrastructure for GSM is available, the use of this technology for V2V communication requires high bandwidth. RFID based V2V communication do not require line of sight. RFID based traffic control can detect emergency vehicles if it is in the frequency range of the traffic control. The prime disadvantage of RFID based V2V communication is low security offered where RFID messages are prone to unauthorized access. The metallic chassis of the vehicle can also cause problems to the RFID transceiver. IR based emergency vehicle detection is Unreliable as IR signals require line of sight and sunlight can cause interference with IR signals. ZigBee based traffic management is a good alternative when compared to other communication protocols mentioned above. It offers wide range of communication does not require line of sight and is comparatively economical. However, to implement V2V communication or traffic management using ZigBee involves restrictions on the number of nodes per controller. Various research papers have proposed the use of Li-Fi for emergency vehicle based traffic control systems. It does not need any infrastructure like GSM. It offers a much secure communication compared to RFID. It is relatively economical compared to ZigBee. It does not use the congested 2.4 GHz frequency like ZigBee. In this paper, Li-Fi based V2V communication system is implemented for smart control of traffic signal.

Chapter 3

PROBLEM IDENTIFICATION

We cannot predict the impact of emergencies like fires,earthquakes, floods, and medical emergencies, among others,over the people that suffer. However, what we can do is try to make these awful times as less traumatic as possible. Fast attention to the scene in the case of a natural disaster or a medical emergency can lower the affections the victims may have later. We can improve the response time by implementing an Emergency Response System (ERS) according to the city's conditions. Knowing the conditions, geography, and demography of a town, or sections of it, allows better planning for this kind of systems. Many problems can arise in cities with a population continually growing but with the same infrastructure. It can only cause more traffic going through the same points. In this scenario, it is tough to follow the 8-10 minutes response time recommendation for medical emergencies like a cardiac arrest or a stroke.

As a result, there are specific critical points in every city where the traffic slows, and it is impossible to take another path to avoid it. This situation makes the emergency responsetime to be longer than it should be. The objective is to turn this system from an adjustable one to a dynamic one where the emergency services have priority without collapsing the city's traffic flow.

Fig. 3.1 Ambulance stuck in traffic jam

Chapter 4

OBJECTIVES

The main objective of the proposed project is to find a solution for the emergency vehicles which get stuck in the traffic jam and could not reach the destination in time.

The features that we are implementing in this project are as follows:

- To provide maximum communication for ambulance from environment and hospitals so that it can make various decisions.
- To provide immediate response such as bed available or not from the hospital side for emergency vehicle.
- To connect all hospital to one platform using IOT so that hospitals at different locations can respond to nearest ambulance.
- The ambulance get the near by hospital indication availability of emergency or in service for patient.
- To provide the traffic clearance for emergency vehicles ambulance by switching the traffic light as per the situation.

Chapter 5

METHODOLOGY and TOOLS

In the proposed methodology, we make use of three subject:

- LIFI- it is the data transmission method to transmit data from ambulance to signal and from signal to web portal and vice versa.
- Arduino Uno- it plays the role of controller in the proposed system. It recieves signal as the input and directs other appliances on basis of signal.
- Adafruit IO- it is the tool used to manage the data on the cloud and helps in running the web portal.

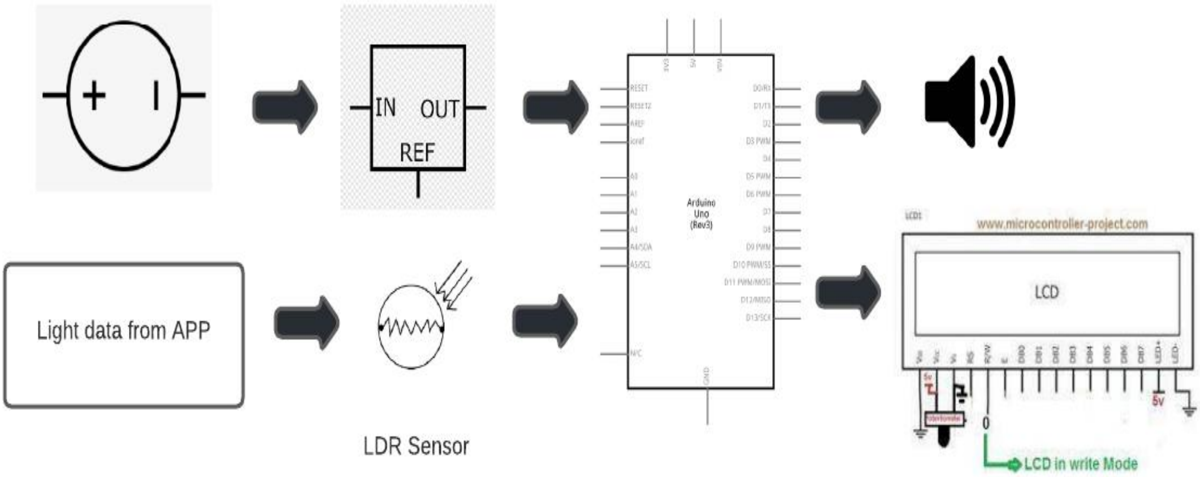

Fig 5.1 LIFI and Arduino UNO interaction

1. LIFI

Li-Fi (also written as LiFi) is a wireless communication technology which utilizes light to transmit data and position between devices. The term was first introduced by Harald Haas during a 2011 TED Global talk in Edinburgh.

Li-Fi is a light communication system that is capable of transmitting data at high speeds over the visible light, ultraviolet, and infrared spectrums. In its present state, only LED lamps can be used for the transmission of data in visible light.

In terms of its end user, the technology is similar to Wi-Fi — the key technical difference being that Wi-Fi uses radio frequency to induce a voltage in an antenna to transmit data, whereas Li-Fi uses the modulation of light intensity to transmit data. Li-Fi is able to function in areas otherwise susceptible to electromagnetic interference (e.g. aircraft cabins, hospitals, or the military).

Bg-Fi is a Li-Fi system consisting of an application for a mobile device, and a simple consumer product, like an IoT (Internet of Things) device, with color sensor, microcontroller, and embedded software. Light from the mobile device display communicates to the color sensor on the consumer product, which converts the light into digital information. Light-emitting diodes enable the consumer product to communicate synchronously with the mobile device.

Li-Fi is a derivative of optical wireless communications (OWC) technology, which uses light from light-emitting diodes (LEDs) as a medium to deliver network, mobile, high-speed communication in a similar manner to Wi-Fi. The Li-Fi market was projected to have a compound annual growth rate of 82% from 2013 to 2018 and to be worth over $6 billion per year by 2018. However, the market has not developed as such and Li-Fi remains with a niche market.

Visible light communications (VLC) works by switching the current to the LEDs off and on at a very high speed, beyond the human eye's ability to notice. Technologies that allow roaming between various Li-Fi cells, also known as handover, may allow to seamlessly transition between Li-Fi. The light waves cannot penetrate walls which translates to a much shorter range, and a lower hacking potential, relative to Wi-Fi. Direct line of sight is not always necessary for Li-Fi to transmit a signal and light reflected off walls can achieve 70 Mbit/s.

Li-Fi can potentially be useful in electromagnetic sensitive areas without causing electromagnetic interference. Both Wi-Fi and Li-Fi transmit data over the electromagnetic spectrum, but whereas Wi-Fi utilizes radio waves, Li-Fi uses visible, ultraviolet, and infrared light. Researchers have reached data rates of over 224 Gbit/s, which was much faster than typical fast broadband in 2013. Li-Fi is expected to be ten times cheaper than Wi-Fi. The first commercially available Li-Fi system was presented at the 2014 Mobile World Congress in Barcelona.

2. ARDUINO UNO

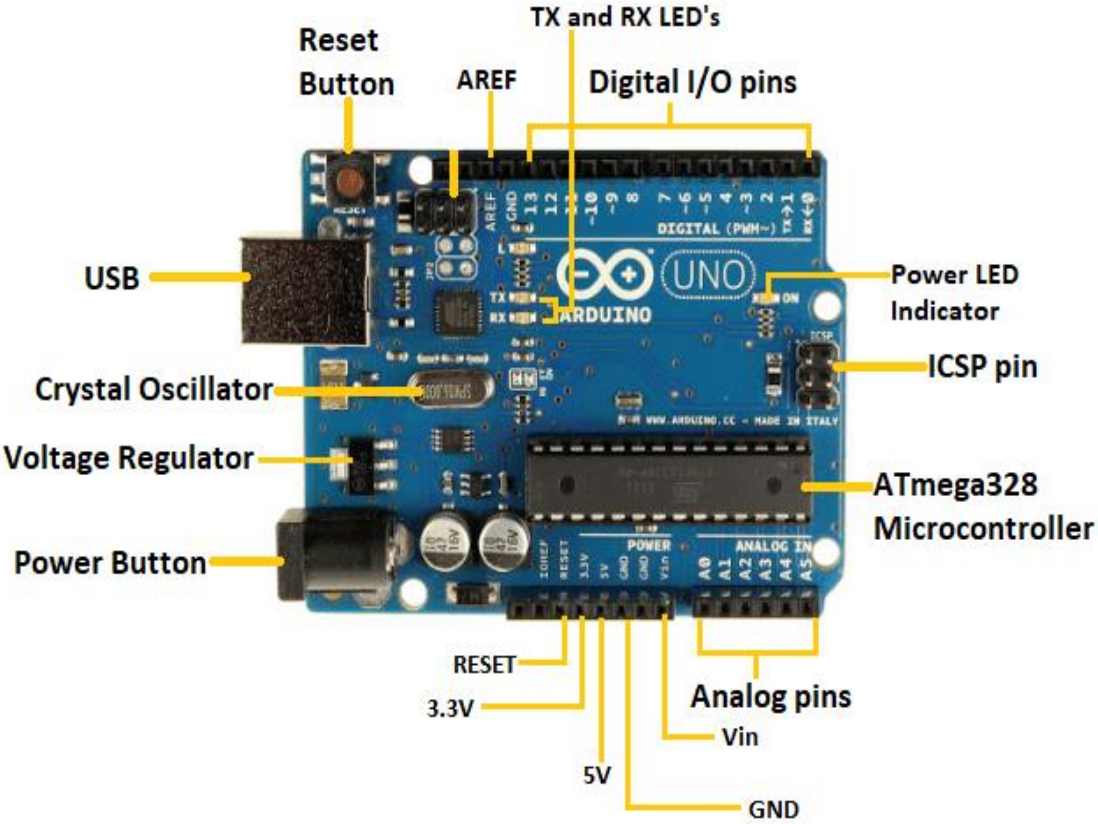

Fig 5.2 Arduino UNO pin diagram

The Arduino UNO is a standard board of Arduino. Here UNO means 'one' in Italian. It was named as UNO to label the first release of Arduino Software. It was also the first USB board released by Arduino. It is considered as the powerful board used in various projects. Arduino.cc developed the Arduino UNO board.

Arduino UNO is based on an ATmega328P microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits.

The Arduino UNO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on both online and offline platforms.

3. Adafruit IO

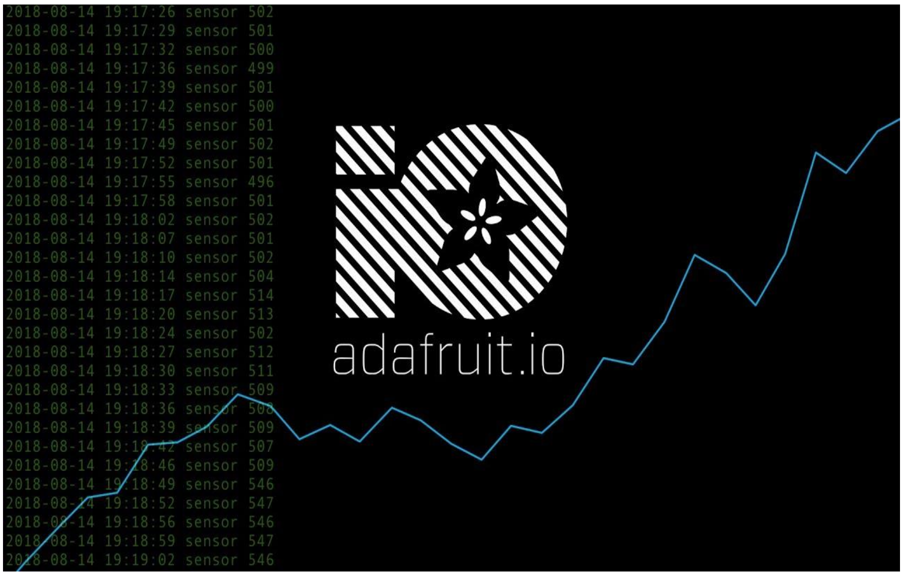

Fig 5.3 Adafruit.IO logo

With the rise in digital transformations, IoT deployments in the cloud have become more popular. By deploying IoT solutions on the cloud, we have the following benefits:

- Cost - Reduces the cost of computing and storage by using various cloud services.
- Scalability - The “pay-as-you-go” pricing model allows a flexible pay model, also allowing scalability of the application.
- Data control - Data backup and recovery with high security.
- Server uptime - Allows very minimum or no downtime, with high server availability.

Adafruit IO is one such cloud provider focusing more on IoT deployments on the cloud. Adafruit IO supports different hardware like Raspberry PI, ESP2866, and Arduino.IoT developers prefer Adafruit IO over other IoT cloud providers for the following reasons:

- Powerful API - Provides us libraries for various programming languages, which also provides the built-in user interface support.
- Dashboard - Understanding data via charts and graphs enables us to make better decisions.
- Privacy - Data is secured in the cloud platform with better encryption algorithms.

Chapter 6

SYSTEM DESIGN

6.1 System Requirements

System requirement specifications gathered by extracting the appropriate information to implement the system. It is the elaborative conditions which the system need to attain. Moreover, the SRS delivers a complete knowledge of the system to understand what this project is going to achieve without any constraints on how to achieve this goal. This SRS not providing the information to outside characters but it hides the plan and gives little implementation details.A good system is the one in which

- System should do minimal computations on its own
- Cost of Implementation should be low
- System should be reliable
- System should be flexible for future enhancements
- System should be Easily Implementable

Hardware Requirement

- Controller(Arduino Uno)
- LASER OR LED
- Phototransistor
- LCD DISPLAY
- WIFI
- LEDs

Software Requirement

- ARDUINO IDE
- ADAFRUIT IO(WEB application)

6.2 System Design

The proposed system includes three modules:

1. Ambulance: Using the LIFI techonology for the ambulances, it becomes very easy for the vehicle to get a way at the traffic. The data is sent continuosly from the laser, the reciever is at the signal.

2. Traffic Signal : The traffic signal recieves the data and clears the path and the data is sent to IOT platform to communicate with hospitals indicating that the ambulance is at particular area.

3. Web Portal: All the hospital are connected on single IOT platform. The hospital which is nearer and available can send the message to the ambulance.

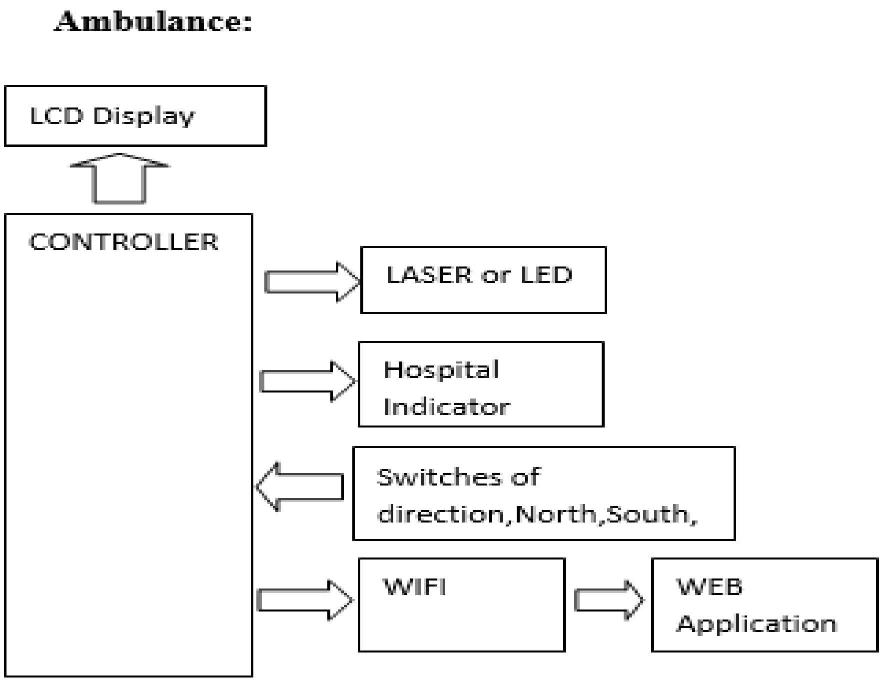

Fig 6.1 Ambulance Module

Ambulance will have the following features:

- It will be mounted with laser or led to send light transmission from ambulance to traffic light.
- It will have lcd display to display the message received from controller of traffic signal.
- It will receive commands to change the direction.
- It will be connected with wifi for connection with the web application.

Traffic:

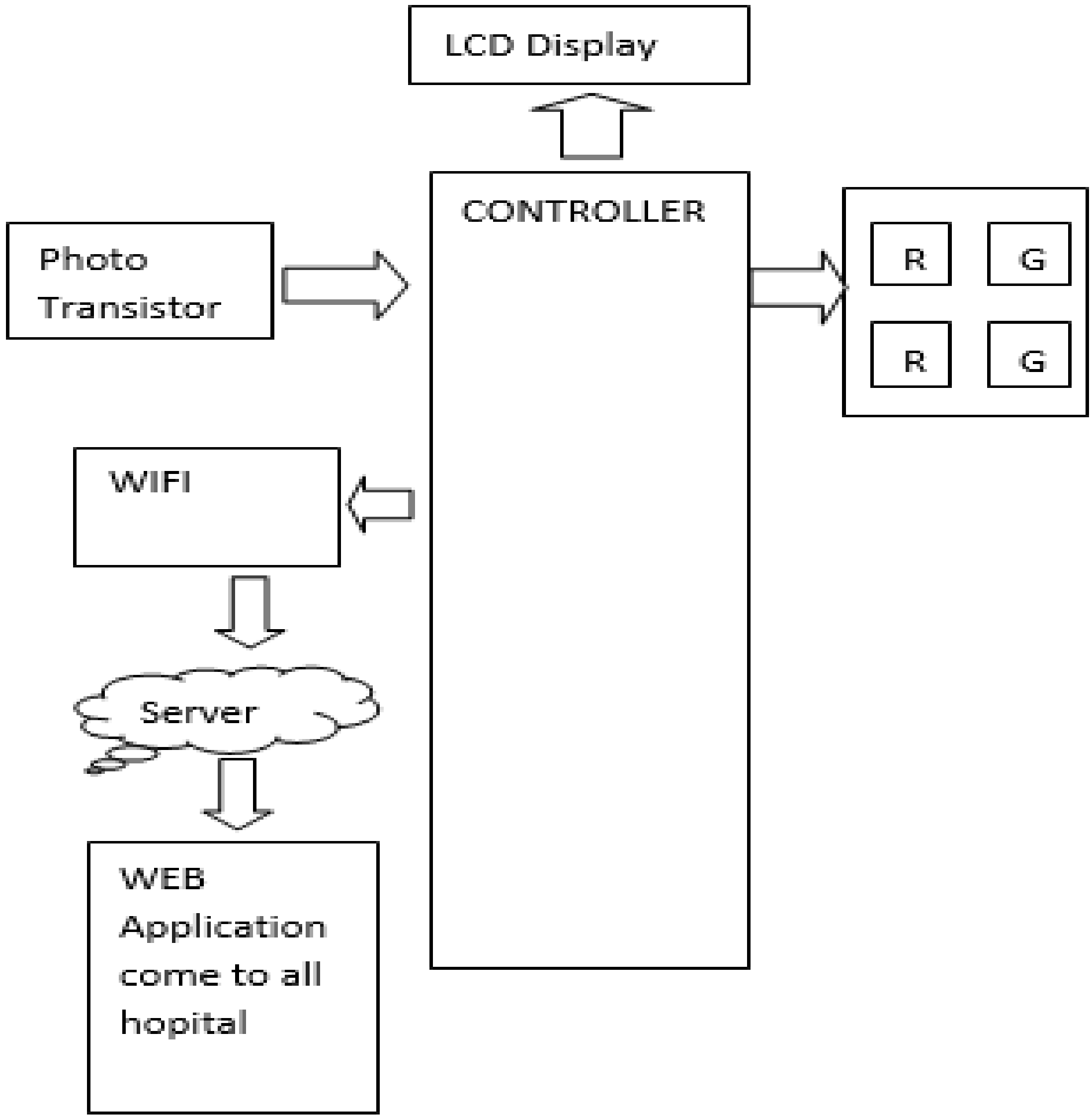

Fig 6.2 Traffic Module

Traffic light will have the following features:

- It will receive the data through the photo transistor from the laser/led of the ambulance.
- It has lcd display to display the message.
- It will change the traffic light as per the situation.
- It will be connected with wifi for receiving and sending data to the web application.

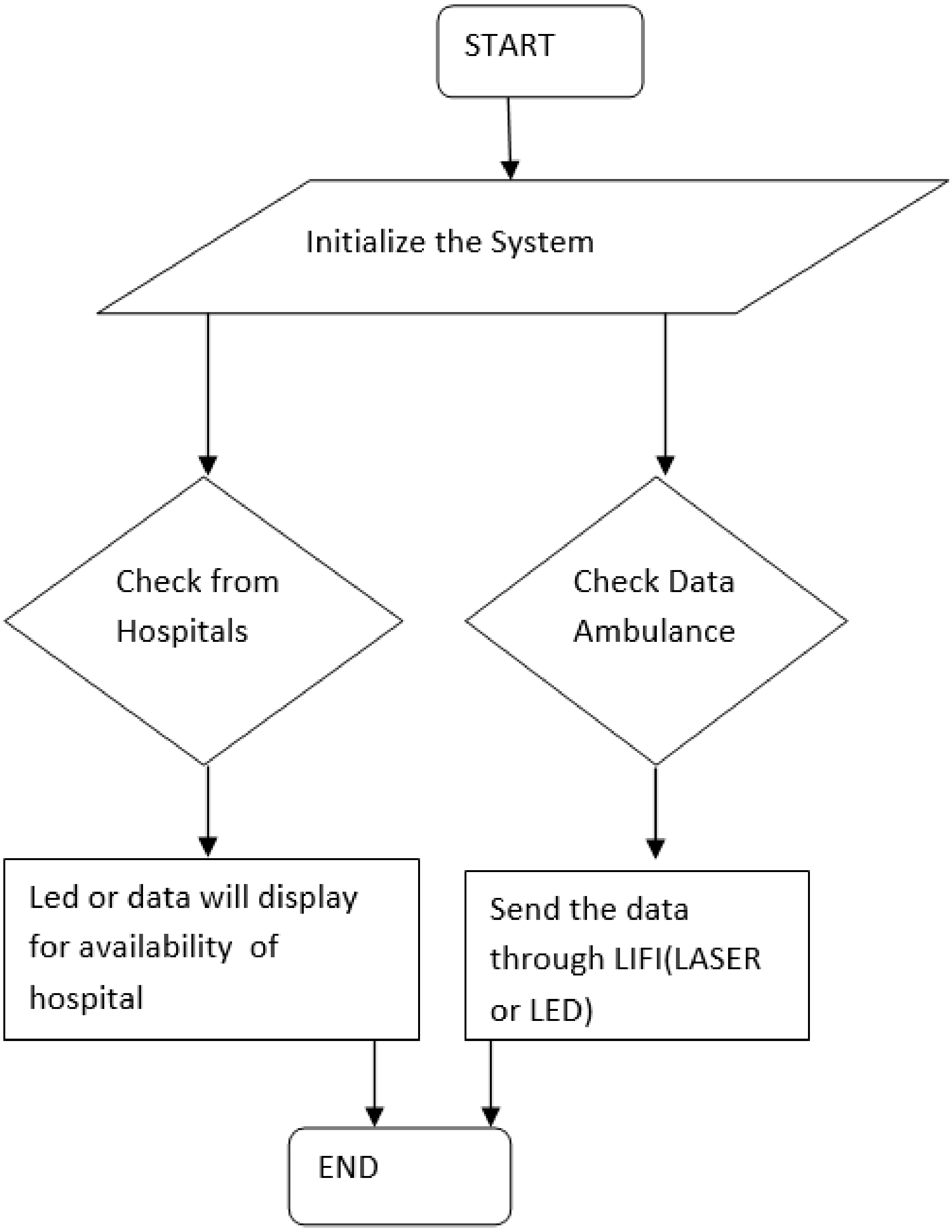

Fig 6.3 Web portal module

Web application will have the following features:

- It will first initialize the system.
- It will check all the data from both hospital and ambulance
- Led will display for availability of hospital
- The data will be send through LIFI

Chapter 7

CONCLUSION

- Aim of the project is to guide an ambulance to the nearest hospital where bed is available in the best way possible.
- This project presents an implementation of automatic traffic light control using the Lifi technology for transmission of data.
- Additionally, the traffic light are powered through solar panel which is nature friendly.
- Since the lifi system is used, the transmission of data happens in visible light spectrum which is the need of the hour.
- All the real time scenario of the traffic management was taken into consideration.
- Given the right guidance and support its application and availability can be enhanced.

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